CHAPTER 3

METHODOLOGY

In this chapter, the methodology will be discussed the strategies and procedures used in conducting the study.

3.1 Research Design

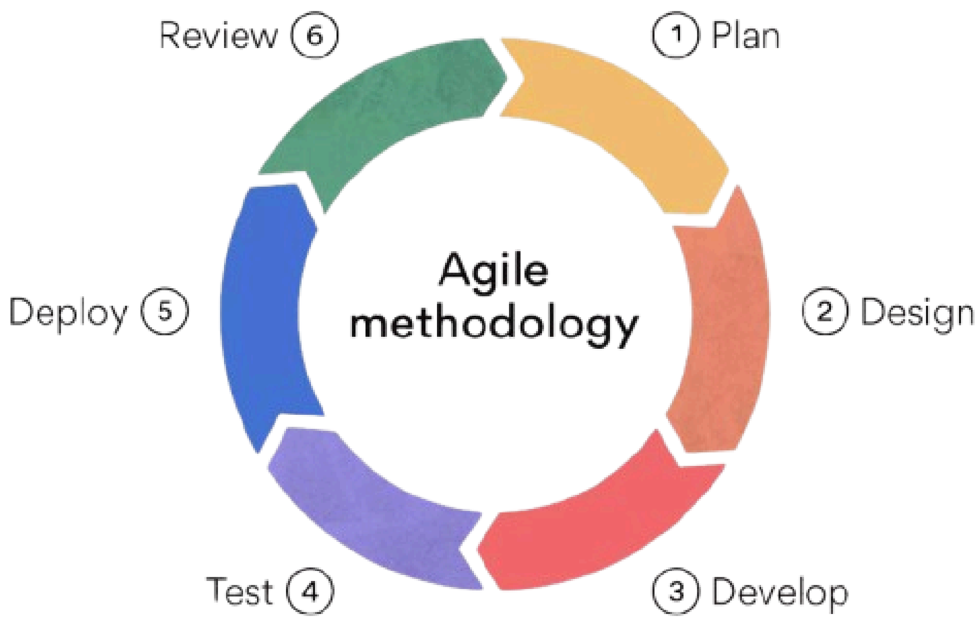
3.1.1 Type of Research

To fully comprehend the study, the researchers employed a developmental and basic quantitative research approach. Developmental research was utilized to systematically assess and refine the eSAS system's design and functionality, focusing on practical application and iterative improvement. This method allowed for continuous feedback and adjustments, ensuring the system's features and performance were effectively aligned with user needs (Richey & Klein, 2007).

. Quantitative research analyzed numerical data, including engagement metrics and performance statistics, to objectively measure the system's effectiveness. This integration of methods offered a well-rounded perspective on the eSAS system's impact and user satisfaction.

3.2 Research Design Approach

The researchers employed the Agile methodology for the study as it aligned perfectly with the nature of the research and effectively addressed the objectives. Following the six fundamental steps of planning, designing, development, testing, deployment and review. The Agile approach allowed the study to advance through the research process, ensuring continuous feedback, adaptability, and timely delivery of outcomes.



Source: (Asana 2023)

Figure 1. **Agile Methodology**

Planning Phase

In the planning phase, the researchers began by conducting an interview with the Student Affairs and Services (SAS) admin to gather initial information about current issues or problems related to student organizations and their management. This step aimed to understand the existing challenges and needs. Based on the identified issues, a solution was formulated, proposing the creation of a web-based system for student organizations, with a focus on enhancing club management and visibility. Following this, a request letter was drafted and submitted to the SAS admin, seeking formal approval to proceed with the project (See Appendix). Once the SAS admin agreed to the partnership, the researchers collected relevant secondhand data, such as existing records and reports, to support the development of the web-based system. Finally, a Gantt chart was created to outline the project's timeline, tasks, and milestones, guiding the process and ensuring efficient progress throughout the study (See Appendix).

Design

In the design phase, the process includes creating an Entity-Relationship Diagram (ERD) to outline the database structure, a Data Flow Diagram (DFD) Level 1 to show data flow and processes, and a Use Case Diagram to illustrate user interactions and system functionality. Additionally, the system architecture will be defined to provide a high-level overview of the system's components and their interactions, including the client-server model, database server, and application server. This comprehensive documentation ensures that the system is well-organized, meets user needs, and effectively manages student organizations and memberships.

Develop

In this phase, the researchers used Visual Studio Code 2019 to develop the system. The icons were sourced from the Font Awesome website and the charts were integrated using CDN data tables with a combination of JavaScript and JQuery to ensure a visually appealing display.

Implementation, coding or development

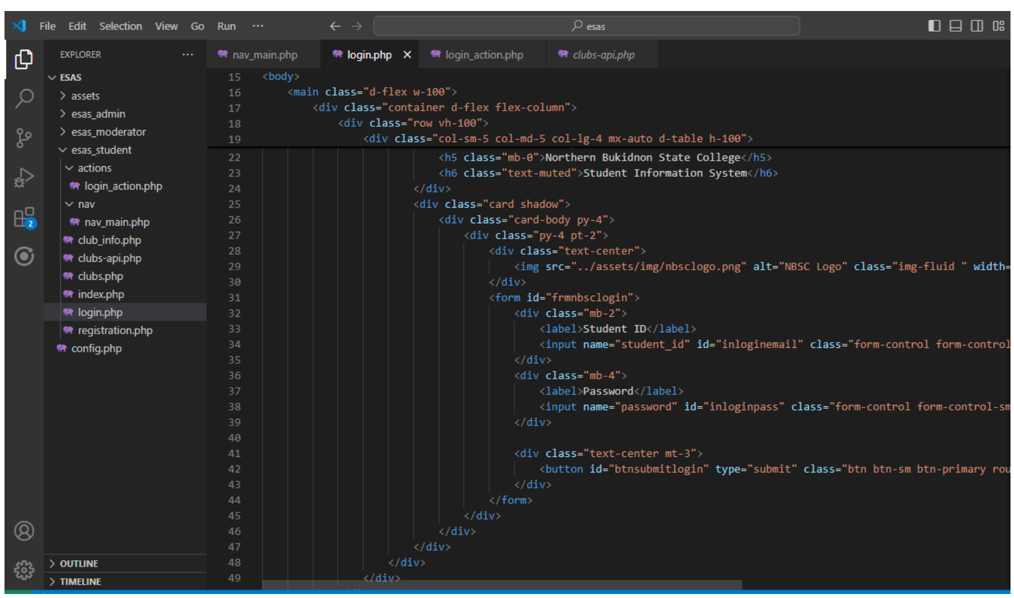


Figure 2. **Code Snippet of System**

During the development stage, the researchers opted for Visual Studio Code as their primary framework. They employed PHP as the main backend language, while for making the layout and interface, they use Bootstrap, complemented by the versatility of JavaScript and the functionality of JQuery, the researchers also used the mySql and Xampp server. An example php file in the Visual Studio Code is shown in the figure above.

Testing

In the testing phase, the system's functionality is rigorously evaluated to ensure it operates correctly and meets specified requirements. The researchers conduct thorough testing to identify and resolve any errors or issues. This includes User Acceptance Testing (UAT), where end-users test the system to confirm that it aligns with their needs and expectations. UAT ensures that the system is user-friendly and meets practical requirements before full deployment. This comprehensive testing approach helps ensure the system's reliability, effectiveness, and overall user satisfaction.

Deployment

Finally, the researchers present their final output to the course instructor and panel for approval through presentation and system demonstration. A PowerPoint was used for the presentation to show the properties and functions of the system. The eSAS system was deployed on NBSC- SAS office for the school's infrastructure. To evaluate the system’s quality the researchers used ISO 25010:2011, this will identify the quality features of the system.

3.3 Hardware and Software Specification

The following are the hardware and software specifications of the project. These were requirements for the implementation and design of the application.

3.3.1 Hardware Specification for Development

The table below reflects the hardware components of the project that will be used during the development phase.

Table 1. **Hardware Development Specification**

HARDWARE (LAPTOP)	SPECIFICATION
Processor	Atleast Intel(R) Core(TM) 13-7100 @ 3.90GHz. (4 CPUs), ~3.9GHz
Installed RAM	At least DDR3 8GB RAM
System Type	64-bit Operating System, x64-based processor

3.3.2 Software Specification for Development

Table 2 discusses the software specification that will be used to develop the system.

Table 2. **Software Development Specification**

Software	**Specification**
Operating System	Windows 10
WampServer	2.4x86

3.3.3. Hardware Specification for Deployment

Table 3 discusses the hardware specifications that will be used during the systems' deployment stage.

HARDWARE	**SPECIFICATION**
Processor	At least AMD A6-7480 Radeon R5, 8 Compute Cores 2C+6G 3.50 GHz or higher
Installed RAM	At least 8.00GB or higher
System Type	64-bit Operating System, x64-based processor

3.3.4. Software Specification for Deployment

Table 4 discusses the software specifications that will be used during the systems' deployment stage.

Table 4. **Software Specification for Deployment**

SOFTWARE	SPECIFICATION
XAMPP	v3.3.0
Google Chrome Microsoft Edge Brave	V 120.0.6099.71 (64-bit) V 120.0.2210.121 (64-bit) V 1.60.125 (64-bit)
Operating System	Windows 10 v21H3

3.5 Evaluation Tool

The evaluation portion is to determine the feedback of the users and keep assured on the quality of the final output. ISO/IEC 25010 is used for the system evaluation given to the participants. The proponents chose two (2) respondents to test the application and gathered their feedback by letting the respondents answer the prepared research instrument.

Formula:

$$\frac{Total\ Average}{\text{numbers of questioned item}} = Overall\ Average$$

Table 3. **ISO/IEC 25010 Survey Test**

Name:____ **Position:** ________________________

CHARACTERISTICS	SUB-CHARACTERISTICS	QUESTIONS	POOR 1	FAIR 2	GOOD 3	VERRY GOOD 4	EXCELLENT 5
Functionality Suitability	Completeness	Does the system have complete features?					
	Correctness	Does the system provide correct information and function appropriately when used?					
	Appropriateness	Are the systems functionality and features appropriate to use?					
Performance Efficiency	Time- Behavior	Does the system perform the task in a timely manner?					
	Capability	Does the system meet the expected requirements?					
	Resource Utilization	Does the amount and type of resources of the system when performing functions meet requirements?					
Compatibility	Co-Existence	Does the system perform functions without affecting other records or components?					
	Interoperability	Does the system allow one to exchange information from one device to another and use that information?					
Usability	Appropriateness Recognizability	Is the system appropriate for your needs?					
	Learnability	Is the system easy to learn?					
	Operability	Is the system easy to use?					
	Use Error Protection	Does the system design appropriate and user-friendly?					
	User Interface Aesthetics	Is the system design appropriate and user-friendly?					
	Accessibility	Is it possible with a diverse set of characteristics and abilities to use the system?					
Reliability	Maturity	Does the system meet the needs for reliability under normal operation?					
	Availability	Is the system reliable and convenient to use when needed?					
	Fault Tolerance	Does the system still function despite encountering some errors when used?					
	Recoverability	Can the system recover data in the event of a failure?					
Security	Confidentiality	Does the system protect user information and only allow authorized people to use it?					
	Integrity	Does the system prevent unauthorized access?					
	Non- Repudiation	Does the system provide information that an action has taken place?					
	Accountability	Can the system track entity actions?					
	Authenticity	Does the system require authentication that a user is really the authorized person accessing the system code?					
Maintainability	Modularity	Do the changes in the system components not affect other components?					
	Reusability	Can the system be used by different department and agencies?					
	Analyzability	Does the system provide analytical reports such as errors and records?					
	Modifiability	Can the system easily modified without affecting other components?					
	Testability	Does the system effectively and efficiently meet the test criteria?					
Portability	Adaptability	Does the system have the capacity to adapt to changes?					
	Installability	Can the system be easily installed and uninstalled?					
	Replaceability	Can the system be easily replaced by a new system in the same environment?					

Table 6 shows the ISO/IEC 25010 Survey Test that was used during the evaluation of the system.

3.6.2 Distribution of Respondents

Table 7. **Distribution of Respondents**

Respondents	**No.**	**Percentage**
Total:		

Table 7 shows the total number of respondents.